

Poisson process

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Definitions I

Recall that an integer-valued random variable X is said to follow the **Poisson distribution** with parameter/mean $\lambda (\geq 0)$ if its distribution is Poisson with parameter $\lambda \geq 0$, that is, if $P(X = k) = \lambda^k e^{-\lambda} / k!$, $k = 0, 1, \dots$

Also recall that a random variable is said to follow the **exponential distribution** with rate $\lambda > 0$ if its distribution has a density with respect to the Lebesgue measure given by $\lambda e^{-\lambda x}$, $x \geq 0$.

Moreover, a random variable is said to follow the **gamma distribution** with shape parameter $k (> 0)$ and rate parameter $\lambda (> 0)$ if its distribution has a density with respect to the Lebesgue measure given by $\Gamma(k)^{-1} \lambda^k x^{k-1} e^{-\lambda x}$, $x \geq 0$.

Finally note that if ξ_i , $i = 1, 2, \dots$ are i.i.d. exponential with rate λ , then $\sum_{i=1}^n \xi_i$ is gamma with shape parameter n and rate parameter λ , for any $n = 1, 2, \dots$

Definitions II

In the simplest form, a Poisson process is defined to be a stochastic process with almost surely càdlàg sample paths and independent and stationary Poisson distributed increments. Formally, a stochastic process N on $\mathbb{R}_+$ with state space $\mathbb{Z}_+$ is called a **Poisson process** with rate/intensity $\lambda \geq 0$ if

- i. The sample paths of N is almost surely càdlàg (i.e. right continuous with left limits);
- ii. (Independent increments) For any $0 \leq t \leq u$,
 $N_u - N_t \perp\!\!\!\perp \mathcal{F}_t = \sigma\{N_s; s \leq t\}$.
- iii. (Stationary increments) For any $0 \leq s \leq t$, $N_t - N_s \stackrel{d}{=} N_{t-s}$.
- iv. For any $t \geq 0$, N_t follows a Poisson distribution with mean λt .

Alternative definition I

We call a stochastic process on $\mathbb{R}_+$ a **counting process** if its state space is the set of nonnegative integers and its sample path is nondecreasing and càdlàg. Poisson process is clearly a special counting process. The following result can be considered as an alternative definition of the Poisson process.

Proposition 1

A counting process N is a Poisson process, iff

1. $N_0 = 0$ a.s.
2. N has stationary and independent increments.
3. $P(N_h = 1) = \lambda h + o(h)$, $h \downarrow 0$.
4. $P(N_h \geq 2) = o(h)$, $h \downarrow 0$.

Alternative definition II

Proof.

The “only if” part is straightforward. We only show the “if” part. To do this, let $p_n(t) = P\{N_t = n\}$. By the stationary and independent increments property and the assumed condition,

$$p_0(t+h) = p_0(t)p_0(h) = p_0(t)(1 - \lambda h + o(h)), \quad h \downarrow 0$$

and hence $\frac{d}{dt}p_0(t) = -\lambda p_0(t)$. This together with $p_0(0) = 1$ implies

$$p_0(t) = e^{-\lambda t}.$$

Similarly, for $n \geq 1$, we have $\frac{d}{dt}p_n(t) = -\lambda p_n(t) + \lambda p_{n-1}(t)$ or $\frac{d}{dt}\{e^{\lambda t}p_n(t)\} = \lambda e^{\lambda t}p_{n-1}(t)$. This together with $p_1(0) = 0$ implies $p_1(t) = \lambda t e^{-\lambda t}$. Then by induction, we get $p_n(t) = (\lambda t)^n e^{-\lambda t} / n!$. □

Waiting times between jumps I

The sample paths of a counting process (a.s.) start from 0, and are piecewise constant, nondecreasing, and right continuous, with jumps of size 1. If we let $\tau_0 = 0$, and let $0 < \tau_1 < \tau_2 < \dots$ denote the 1st, 2nd, $\dots$, jump times of the Poisson process, and use X_i , $i = 1, 2, \dots$ to denote the waiting times between successive jumps, that is, $X_i = \tau_i - \tau_{i-1}$, then from $\tau_n = \inf\{t > \tau_{n-1}; N_t > N_{t-}\}$ it is easy to see that the τ_n are stopping times, and that

$$\{\tau_n \leq t\} = \{N_t \geq n\}. \quad (1)$$

The following result gives the distributions of the jump times and the waiting times.

Waiting times between jumps II

Proposition 2 (Distributions of the jump times and the waiting times)

Let N be a Poisson process. Let τ_i , $i = 0, 1, 2, \dots$ and X_i , $i = 1, 2, \dots$ be defined as above. Then the distribution of τ_n is gamma with shape parameter n and rate parameter λ . The X_i , $i = 1, 2, \dots$ are independently and identically distributed according to an exponential distribution with rate parameter λ .

Proof.

By (1) and the independent increments property, it follows from straightforward calculations that

$\frac{d}{dt}P\{\tau_n \leq t\} = (n-1)!^{-1}\lambda^n t^{n-1}e^{-\lambda t}$, $P\{X_1 > t\} = e^{-\lambda t}$, and $P\{X_n > t | X_1 = s_1, \dots, X_{n-1} = s_{n-1}\} = e^{-\lambda t}$ for all n , and all $s_1, \dots, s_n, t > 0$. □

Construction of Poisson Process I

As a counting process, the Poisson process can serve as a model for counting the number of events happening at random times. If we let X_1 denote the waiting time from time 0 until the 1st event, X_i the waiting time from the $(i - 1)$ st event to the i th event, $i = 2, 3, \dots$. Define $\tau_n = \sum_{i=1}^n X_i$, $n = 0, 1, \dots$, and $N_t = \#\{i; \tau_i \leq t\} = \sup\{i; \tau_i \leq t\}$. That is, τ_n is the arrival time of the n th event, and N_t counts the number of event that have happened by time t . The following result says the converse of Proposition 2 is also true. It also gives a simple method of perfect simulation of a Poisson process.

Theorem 3

Let X_i , $i = 1, 2, \dots$ be i.i.d. exponential random variables with rate $\lambda > 0$. Let τ_i , $i = 0, 1, \dots$ and N_t , $t \geq 0$ be defined as in the preceding paragraph. Then N is a Poisson process with rate λ .

Proof.

Construction of Poisson Process II

That $N_0 = 0$ and N_t is cádlág as a function of t follows from the definition. To see that N_t is Poisson with mean λt , we note τ_n is gamma with shape n and rate λ , and therefore by (1) and integration by parts,

$$\begin{aligned}P\{N_t = n\} &= P\{N_t \geq n\} - P\{N_t \geq n+1\} = P\{\tau_{n+1} > t\} - P\{\tau_n > t\} \\&= (n!)^{-1} \int_t^\infty \lambda^{n+1} x^n e^{-\lambda x} dx - \{(n-1)!\}^{-1} \int_t^\infty \lambda^n x^{n-1} e^{-\lambda x} dx \\&= \lambda^n t^n e^{-\lambda t} / n!.\end{aligned}$$

To show the independent increments property, let

$$N_r^* = \#\{i; \tau_{N_s+i} - s \leq r\}, \quad r \geq 0.$$

It is easy to see that $N_{t-s}^* = N_t - N_s$ for all $s < t$. From

$$\begin{aligned}P\{\tau_{N_s+1} - s > x, N_s = k\} &= P\{\tau_{k+1} - s > x, \tau_k \leq s < \tau_k + X_{k+1}\} \\&= \int_0^s \Gamma(k)^{-1} \lambda^k u^{k-1} e^{-\lambda u} e^{-\lambda(s+x-u)} du = k!^{-1} \lambda^k s^k e^{-\lambda(s+x)} \\&= P\{N_s = k\} e^{-\lambda x},\end{aligned}$$

we note that $\tau_{N_s+1} - s$ is exponential with rate λ independent of N_s .

Construction of Poisson Process III

Proof.

Similarly we see that $\tau_{N_s+2} - \tau_{N_s+1}$ is exponential with rate λ and is independent of $(N_s, \tau_{N_s+1} - s)$ and that $(\tau_{N_s+i+1} - \tau_{N_s+i})$ is exponential with rate λ and is independent of $(N_s, \tau_{N_s+i-1} - s, \dots, \tau_{N_s+i} - \tau_{N_s+i-1})$ for all $i \geq 2$. Therefore, the process N^* is independent of N_s and has the same distribution as N . It follows that $N_{t-s}^* = N_t - N_s$ is Poisson distributed with mean $\lambda(t-s)$, which shows the stationary increments property.

To complete the proof of the independent increments property, let $0 < t_1 < \dots < t_n$ be arbitrary. By the previous argument, we have $N_{t_n} - N_{t_{n-1}} \perp\!\!\!\perp N_{t_{n-1}} - N_{t_{n-2}}$ by interpreting both random variables as increments of the process

$N_t^{t_{n-2}} = N_{t_{n-2}+t} - N_{t_{n-2}}, t \geq 0$. Similarly we have $(N_{t_n} - N_{t_{n-1}}, N_{t_{n-1}} - N_{t_{n-2}}) \perp\!\!\!\perp N_{t_{n-2}} - N_{t_{n-3}}$, etc. This shows the independent increments property and therefore completes the proof of the result. □

Uniformity of jump times I

Recall that for i.i.d. random variables $X_1, \dots, X_n$, their i th order statistic $X_{(i)}$ is defined to be the i th smallest value among $X_1, \dots, X_n$, so for instance, $X_{(1)} = \min\{X_1, \dots, X_n\}$ and $X_{(n)} = \max\{X_1, \dots, X_n\}$. If $X_1, \dots, X_n$ are continuous variables and their common density function is f , then the joint distribution of the order statistics has a density given by

$$n! \prod_{i=1}^n f(x_i) 1_{\{x_1 \leq \dots \leq x_n\}}.$$

The following result gives an efficient method to simulate a Poisson process up to a given time point t .

Theorem 4

Given $N_t = n$, the jump times $\tau_1, \dots, \tau_n$ have the same distribution as the order statistics corresponding to n independent random variables uniformly distributed on the interval $(0, t)$.

Uniformity of jump times II

Proof.

Show that the conditional density function of $\tau_1, \dots, \tau_n$ given $N(t) = n$ is

$$f(t_1, \dots, t_n) = \frac{n!}{t^n} 1_{\{0 < t_1 < \dots < t_n\}}.$$



By this result, to simulate a Poisson process with rate λ on $[0, t]$ we just need to generate a Poisson variable N with mean λt and then generate N i.i.d. $U(0, t)$ variables $U_1, \dots, U_N$. Then their order statistics $U_{(1)}, \dots, U_{(N)}$ will be the the jump times of the Poisson process.

Non-homogeneous Poisson process I

If we drop the stationary increments property required of the Poisson process then we have a non-homogeneous Poisson process. Formally, a **non-homogeneous Poisson process** with rate/intensity function λ_t , $t \geq 0$ is defined as a counting process N_t , $t \geq 0$ with the following properties,

- (i). $N_0 = 0$.
- (ii). $\{N_t, t \geq 0\}$ has independent increments.
- (iii). $P(N_{t+h} - N_t \geq 2) = o(h)$.
- (iv). $P(N_{t+h} - N_t = 1) = \lambda_t h + o(h)$.

Remark 5

*With this definition, it can be shown using an argument similar to the one used in the proof of Proposition 1 that for all $s, t \geq 0$, $N_{t+s} - N_t$ follows a Poisson distribution with mean $\Lambda_{t+s} - \Lambda_t$, where $\Lambda_t = \int_0^t \lambda_u du$ is known as the **cumulative intensity function**, or **compensator** of the (non-homogeneous) Poisson process.*

Non-homogeneous Poisson process II

Remark 6

It is clear that $N_t - \Lambda_t$, $t \geq 0$ is a martingale with respect to the natural filtration of N_t , $t \geq 0$.

Remark 7

It is a simple exercise to show that given $N_t = n$, the joint distribution of the jump times up to time t is the same as that of the order statistics of n i.i.d. random variables with density

$$f(x) = \frac{\lambda(x)}{\Lambda(t)} 1_{\{0 \leq x \leq t\}}.$$

Non-homogeneous Poisson process III

Remark 8

If N is a (non-homogeneous) Poisson process with cumulative intensity function $\Lambda(t)$, $t \geq 0$, then it is straightforward to verify that the counting process $\tilde{N}$ obtained by the following time change is a Poisson process with constant rate 1:

$$\tilde{N}(t) = N(\Lambda^{-1}(t)), \quad t \in [0, \Lambda_{\infty}),$$

where Λ^{-1} is the inverse function of Λ , that is

$$\Lambda^{-1}(t) = \sup\{u; \Lambda(u) \leq t\}.$$

This result is a useful tool for diagnostics of the Poisson process model. After obtaining the fitted intensity $\hat{\lambda}(t)$ to the original data, say $0 < t_1 < t_2 < \dots < t_n \leq T$, calculate the transformed event times: $\tilde{t}_i = \hat{\Lambda}(t_i) = \int_0^{t_i} \hat{\lambda}(x) dx$, $i = 1, \dots, n$, and test the uniformity of $\tilde{t}_i$, $i = 1, \dots, n$ on the interval $[0, \hat{\Lambda}(T)]$.

Thinning and superposition of Poisson processes

If $N(t)$, $t \geq 0$ is a counting process with jump/event times $0 < \tau_1 < \tau_2 < \dots$. Let $p(t)$, $t \geq 0$ be a function taking values in $[0, 1]$. At each jump time τ_i , if the event is deleted with probability $1 - p(\tau_i)$ (or retained with probability $p(\tau_i)$), then the counting process associated with the retained event times is called the $p(\cdot)$ -**thinning** of $N(t)$.

If N_t is a Poisson process with intensity function $\lambda(t)$, then its $p(\cdot)$ -thinning is a Poisson process with intensity function $\lambda(t)p(t)$.

If $N_1(t)$ and $N_2(t)$ are two counting processes, then the counting process with jump times consisting of the jump times of $N_1(t)$ and $N_2(t)$ pooled together is called the **superposition** of $N_1(t)$ and $N_2(t)$.

If $N_1(t)$ and $N_2(t)$ are two independent Poisson processes (so they cannot jump at the same time) with intensity functions $\lambda_1(t)$ and $\lambda_2(t)$ respectively, then their superposition $N(t) = N_1(t) + N_2(t)$ is a Poisson process with intensity function $\lambda_1(t) + \lambda_2(t)$.

Hawkes process I

A **Hawkes process** or a **self-exciting process** is a counting process $N(t)$ with a (stochastic) intensity $\lambda(t)$,

$$\lambda(t) = \lambda_0 + \int_0^{N(t-)} \mu(t-s) dN(s) = \lambda_0 + \sum_{i=1}^{N(t-)} \mu(t - \tau_i),$$

where τ_i , $i = 1, 2, \dots$ denote the jump times of $N(t)$, $\lambda_0 > 0$ is the baseline intensity parameter, and $\mu(t)$ is a positive function such that $\int_0^\infty \mu(t) dt < 1$, called the *excitation function*, or *fertility function*, or *excitation kernel*, or *memory kernel*. The process is called self-exciting because the occurrence of an event will increase the probability of occurrence of future events. The event times of a Hawkes process have stronger clustering (compared with a Poisson process).

Hawkes process II

The Hawkes process is a model for population growth. The jump times (or event times) of the process can be thought of as two types: 'immigrants' without extant parents in the process, and 'offspring' that are produced by existing events.

An construction of the event times is as follows. Immigrants arrive according to a Poisson process at constant intensity λ_0 while the offspring arise as events of Poisson processes with intensity function $\mu(t)$ started at the arrival time of their parent. Individuals (immigrants and offspring) start to produce offspring of their own once they enter the population (by immigration or birth).

Hawkes process III

To fit a Hawkes process, or a general (simple) counting process (also called point processes with points referring to the events or jumps) with a model $\lambda(\cdot)$ for the intensity (stochastic or deterministic), the likelihood function of the process model relative to the observed event times (up to a censoring time T say) is often needed, which is given by

$$L(\lambda) = \prod_{i=1}^{N(T)} \lambda(\tau_i) \exp\left\{-\int_0^T \lambda(t) dt\right\}.$$

To check if a counting process model $N(t)$ fits the observed event times (up to time T): $0 < \tau_1 < \tau_2 < \dots < \tau_n \leq T$, with $n = N(T)$, we can still use the time-change results. That is, if the intensity $\lambda(t)$ is correctly specified for $N(t)$, then the time-rescaled process with event times $\tilde{\tau}_i = \int_0^{\tau_i} \lambda(t) dt$ should be a standard Poisson process (with constant rate 1).

Hawkes process IV

If you are interested in simulating or fitting Hawkes processes (using R), you might like to have a look at the R package IHSEP (<https://cran.r-project.org/package=IHSEP>).